



A peer-to-peer blockchain-based architecture for trusted and reliable agricultural product traceability[☆]

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ABSTRACT

The lack of new and advanced technologies, such as the Internet of Things (IoT) and Fog computing, makes it challenging to trace the origin of agricultural products in supply chains. In addition, the system's traditional centralized architecture cannot provide a trusted traceability service for farming goods in these supply chains. This paper proposes a peer-to-peer Blockchain-based architecture integrating self-sovereign identity (SSI) and a decentralized key management system (DKMS) for a trusted and reliable traceability service for agricultural food products in the supply chain networks. The public Blockchain network Fantom will provide faster and cheaper transactions. In contrast, SSI will ensure the identity of each entity by adopting DKMS technology on top of Blockchain to facilitate faster authenticity, integrity, and confidentiality service of the supply chain's transaction and verification process. The system is deployed on the different public networks, i.e., Fantom and Ethereum, and shows that the proposed Fantom-based Blockchain network outperforms in terms of transaction and verification timing.

1. Introduction

Bangladesh is a predominantly agrarian nation, that heavily depends on its agricultural sector, which employs 50% of the country's workforce [1]. The production of fruits and vegetables in Bangladesh holds significant importance due to their green, nutritious, and healthful qualities, making them widely favored by the population [2]. Bangladesh's agricultural food supply chain involves multiple entities responsible for various aspects, including the production and transportation of farm products. Unfortunately, the current system relies on manual paper records to track and document these processes. This manual approach makes it challenging to trace crucial details like the source, location, quality, and other essential information about agricultural products.

Agricultural product traceability is a vital component of the supply chain, as it enables tracking food product movements and ingredients [3]. It enhances transparency across the supply chain, allowing all stakeholders to verify the quality of agricultural products and easily trace factors such as harvest dates, times, and locations. Traceability also plays a significant role in improving food quality and safety. However, the existing agricultural system falls short of ensuring the

effective traceability of farm products. Recent studies conducted by researchers at Bangladesh Agricultural University have identified the presence of harmful heavy metal elements, including lead, cadmium, and nickel, in eggplants collected directly from farmers in Islampur and Melandaha Upazila of Jamalpur. These heavy metals are known to be carcinogenic and pose a cancer risk to individuals who consume eggplants contaminated with these substances [4]. Consequently, the necessity for agricultural product traceability becomes even more critical, impacting various aspects of daily life.

Numerous strategies have been employed to establish traceability in the agricultural food supply chain. Tasnim et al. [5] demonstrated that farmers rely on middlemen in Bangladesh's conventional agricultural and food supply network due to a lack of effective communication between producers and consumers. They propose a theoretical agri-food supply chain framework using Blockchain technology for Bangladesh. Khan et al. [6] highlighted that the conventional paper-based record-keeping methods in the shrimp supply chain are fragmented, impeding effective traceability and a comprehensive supply chain view. They introduce ShrimpChain, a novel conceptual framework utilizing a hybrid public-private Blockchain to address this issue. Hossain et al. [7] investigated whether Bangladeshi consumers are willing to pay a premium

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for certified fresh foods, finding that consumers are indeed willing to pay more for food safety.

Despite these advancements, current methods fail to provide efficient traceability for farmers and consumers by hindering their ability to track product origins and destinations. Some research uses centralized databases [8] for product information storage and tracking, while others rely on Blockchain [9] as a decentralized database for recording product details. However, there is currently no authentication mechanism for supply chain entities, leading to consumer uncertainty regarding entity authenticity [10]. This problem is exacerbated by the absence of authentication for various actors in the supply chain, resulting in the distribution of spoiled goods by unauthorized sources, thereby jeopardizing public health. Additionally, marginalized farmers and processors face additional obstacles as they lack insights into pricing dynamics involving distributors, retailers, and consumers, limiting their benefits from sales. Moreover, unethical practices, such as hoarding by fraudulent entities, disrupt market availability.

To address these issues, this research proposes an integrated model that combines self-sovereign identity (SSI) and Blockchain technology to establish trustworthy product traceability. By integrating a Decentralized Key Management System (DKMS) and implementing SSI, this model facilitates individual authentication, peer-to-peer communication, confidentiality, integrity, and traceability throughout the agricultural supply chain. Leveraging the faster transaction processing and reduced costs offered by the Fantom Blockchain network, the study introduces a novel packaging and labeling mechanism to ensure product origin, streamlining, and making traceability services more cost-effective. This comprehensive approach enhances transparency, accountability, and the ability to verify the authenticity and origin of agricultural products at every stage of the supply chain. The main contributions of this research are as follows:

- SSI is utilized to enhance entity authentication, enabling quicker and more dependable verification, thereby fostering trust among participants in the supply chain. This is viewed as the initial stride toward attaining traceability for agricultural products.
- A novel approach to packaging and labeling has been suggested to track and confirm the traceability of agricultural products.
- Implementing the Fantom Blockchain network enhances the system's speed, cost-efficiency, scalability, and security.

The remainder of the article is organized as outlined below. Section 2 delves into the examination of related works. Sections 3, 4, and 5 are dedicated to providing details on Preliminaries, System Overview and Setup, and Traceability Service, respectively. Performance Evaluation is presented in Section 6. To wrap up, Section 7 serves as the conclusion for this article.

2. Related work

In recent years, researchers have conducted extensive exploratory studies on agricultural product traceability, contributing to the development of this field. In this section, we will briefly overview their work and discuss the limitations of these novel approaches. Table 1 illustrates the summary of the state-of-the-art techniques with the proposed studies.

Qian et al. [11] implemented a wheat milling traceability system by combining Radio Frequency Identification (RFID) technology and 2D barcodes alongside a centralized database to manage information spanning from raw materials to finished products. Conti [17] introduced a food traceability system that employs smartphones and Near Field Communication (NFC) technology at each supply chain stage, ensuring information delivery to the end consumer. Wang et al. [23] suggested a Hyperledger Fabric Blockchain-based framework for tracking and tracing agricultural food flow in the supply chain. Similarly, YU et al. [19] presented a Blockchain-based framework using Hyperledger Fabric smart contracts for agricultural supply chain traceability.

Several researchers have explored the use of RFID technology, 2D barcodes, smartphones, NFC, Internet of Things (IoT), Quick Response (QR) codes, and centralized databases to enhance agricultural product traceability [12,13].

Salah et al. [16] and Shahid et al. [18] incorporate a decentralized file system such as the Interplanetary File System (IPFS) to achieve traceability and liquidity for food in the agricultural supply chain. Other scholars have combined Blockchain and the IoT to enable real-time tracking of products from farms to dining tables [20]. Dong et al. devised a reliable prototype traceability system for the entire food supply chain by employing an “on-chain + cloud database” data storage mode and leveraging Blockchain technology [15]. Yang et al. [22] introduced a Blockchain-based traceability system for querying and storing agricultural product information. Coco et al. [21] presented a model that combines IPFS, RFID, and Blockchain to establish a visible, secure, and auditable traceability system for Italian food products. Their model enables each supply chain entity to verify product quality. Luisanna et al. [24] proposed a model based on SSI, IPFS, and Blockchain to provide comprehensive visibility of food quality certifications along the supply chain. Hasan et al. [25] proposed a solution integrating Blockchain, IoT, and lightweight asymmetric cryptography, specifically Elliptic Curve Cryptography (ECC), to enhance privacy and security in supply chain transactions. Surasak et al. [14] utilized Blockchain and the IoT to design and develop a traceability system for Thai agriculture products.

Saranya et al. [34] introduced a Blockchain system incorporating user identification and access control mechanisms, addressing the needs of various supply chain participants. Additionally, their system introduced a novel Proof of Transaction (PoTx) consensus algorithm to achieve scalability with reduced communication overhead and computational requirements. Li et al. [38] developed a tripartite evolutionary game model involving producers, supermarkets, and e-commerce platforms. They also examined the dynamic evolutionary process of these three entities and the factors influencing their stable strategy choices. Marchesi et al. [26] proposed a novel approach to customize and assemble Ethereum-based smart contracts tailored for the agri-food industrial sector, emphasizing code and module reuse, process automation, and shortened development times, all while ensuring safety and reliability. Yadav et al. [27] conducted a network analysis on 346 relevant articles from the Web of Science (WoS) database, employing rigorous filtration techniques. Santhiya [39] demonstrated a Blockchain and smart contract-based solution for agriculture-centric Food Supply Chains, where every transaction is securely recorded on the Blockchain and subsequently stored in the Interplanetary File Storage System.

Subashini and Hemavathi et al. [28] conducted a comprehensive survey on agrifood supply chain traceability challenges, exploring multiple Blockchain platforms like Ethereum and Hyperledger in conjunction with IoT techniques. Krishna and Rekha [32] introduced a Food Supply Chain Traceability System using an Ethereum smart contract integrated with an IPFS, capable of handling extensive data volumes. Kukolj and colleagues [35] analyzed Blockchain technology's applications in product traceability within the food supply chain. Katti et al. [29] outlined the potential use of Blockchain and IoT technologies to enhance traceability in the agri-food sector. Wang and team [33] demonstrated the efficiency and security benefits of the dual-chain dual-storage model based on Hyperledger for Blockchain storage queries. Rajput et al. [36] summarized previous research and proposed a traceability system for agricultural supply chains. Subashini and Hemavathi [30] illustrated how Blockchain can effectively manage non-perishable agricultural food supply chains. Zhang et al. [31] introduced a flexible scheme for agricultural product traceability, allowing for dynamic traceability link navigation. Mankawade and coauthors [37] emphasized the critical role of supply chain management in various sectors and the importance of maintaining overall process visibility.

Nevertheless, these methods have drawbacks, including their centralized nature and dependence on traditional databases, which are

Table 1
An overview of existing research on agricultural product traceability.

Study	Year	Technologies			Features	
		IPFS	SSI	Blockchain	Entity authentication	Traceability
[11]	2012	–	–	–	–	✓
[12]	2014	–	–	–	–	✓
[13]	2016	–	–	–	–	✓
[14,15]	2019	–	–	✓	–	✓
[16]	2019	✓	–	✓	–	✓
[17]	2020	–	–	–	–	✓
[18]	2020	✓	–	✓	–	✓
[19,20]	2020	–	–	✓	–	✓
[21–23]	2021	✓	–	✓	–	✓
[24]	2021	✓	✓	✓	–	–
[25]	2022	–	–	✓	✓	–
[26–31]	2022	–	–	✓	–	✓
[32,33]	2022	✓	–	✓	–	✓
[34–37]	2023	–	–	✓	–	✓
[38]	2023	–	–	–	–	✓
[39]	2023	✓	–	✓	–	✓
Proposed	2023	✓	✓	✓	✓	✓

vulnerable to data tampering and the infiltration of false information without detection. Furthermore, the traditional system only assesses traceability at the consumer level, posing difficulties for other supply chain participants in identifying the origins of counterfeit products. Blockchain technology has gained substantial attention from scholars and researchers due to its decentralized, distributed, secure, quicker, transparent, and immutable characteristics. Numerous studies have suggested incorporating Blockchain, IoT, and IPFS to enhance supply chain traceability, as shown in Table 1. However, a common drawback in these investigations is the absence of authentication mechanisms for supply chain participants, compromising traceability's reliability. Moreover, the authentication systems proposed in these studies still require enhancement since they rely on centralized service providers for identity validation. In the prevailing agricultural supply chain, there is no provision for entities to verify their identities autonomously. Conversely, the proposed framework addresses these limitations by delivering completely decentralized authentication for supply chain participants, guaranteeing robust traceability throughout the agricultural supply chain.

3. Preliminaries

This section describes all technologies related to the system. All the notations used in the proposed method are shown in Table 2.

3.1. Blockchain and smart contract

Blockchain technology is characterized by its decentralized and distributed ledger system, where transactions are visible to all network nodes. It enables secure communication and information dissemination among non-trusted members without the need for a central authority. Cryptographic hashes identify each block in the Blockchain and form a chain, with each block containing a hash pointer to the previous block, a timestamp, and a list of transactions [40]. Additionally, a smart contract refers to a computer program or self-executing code that operates on the Blockchain. Its purpose is to enforce the terms of an agreement when specific conditions are met. Smart contracts possess account balances, private storage, and executable code, with their capabilities determined by the programming language used to express the agreement rather than the underlying technology.

The evolution of Blockchain technology can be observed through three distinct generations [41]. The first generation, represented by Bitcoin, aimed to revolutionize existing financial systems by providing

Table 2

Notations.

Sign	Meanings	Sign	Meanings
AD	agricultural department	F	farmer
P K	public key	D	distributor
h()	hash	C	consumer
SC	seed company	H	holder
IPFS	interplanetary file system	V	verifier
SN	serial number	P	processor
DID	decentralized identifier	R	retailer
SK	private Key	CART	carton
VC	verifiable credential	PA	pallet
LC	logistic company	B	box
Comp _{Name}	company name	U	unit
QR	quick response	CA	case
SSI	self-sovereign identity	I	issuer

a decentralized monetary platform that restores individual control. However, it relies on the costly Proof of Work (PoW) consensus algorithm for transaction validation. The second-generation Blockchain introduced smart contracts, enabling direct negotiations between parties and offering a faster and more secure execution of agreements. Yet, scalability and interoperability remained major challenges in both the first and second generations. The lack of scalability results in transaction delays due to high volume and limited space in the Blockchain, while the isolated nature of early Blockchain iterations prevents effective interaction between different Blockchain networks. To address these fundamental flaws, third-generation Blockchains such as Fantom, Cardano, Solana, and Polkadot have emerged [42–45]. These third-generation solutions prioritize scalability and interoperability, enabling Blockchain technology to support mass adoption and overcome issues like slow transaction times and closed systems.

This study leveraged the Fantom Blockchain, an open-source smart contract platform designed for digital assets and decentralized applications (dApps) [46]. Fantom is a layer-1 Blockchain solution that addresses the issues of high costs and slow performance often associated with Ethereum. It utilizes the Asynchronous Byzantine Fault Tolerant (aBFT) Proof-of-Stake (PoS) consensus mechanism called Lachesis. Lachesis operates on a leaderless Directed Acyclic Graph (DAG) model, which is further described in Section 3.4, while the Lachesis

consensus protocol is discussed in Section 3.5. The utilization of Fantom Blockchain enables faster transaction processing and scalability, making it a suitable choice for this study.

3.2. Self-Sovereign Identity (SSI)

Self-Sovereign Identity is the latest approach to digital identity in cyberspace. This innovation permits individuals to own and manage their digital identities without relying on third-party providers to store and centrally maintain the data. The principle objective is to figure out a technique to prove an entity's identity to websites, applications, and services. SSI comprises seven building blocks. They are Verifiable credentials, The trust triangle (issuer, holder, and verifier), Digital wallets, Decentralized identifiers (DIDs), Blockchains or other verifiable data registries, and Governance frameworks [47].

The summary of each building block is described below:

- Verifiable Credentials (VC) are the digital equivalent of physical certifications, which individuals carry in their wallets to prove their identity.
- The trust triangle involves three roles: issuer, holder, and verifier. The issuer is the source of credentials; the holder holds them in his/her wallet, and the verifier checks them when the holder presents them to the verifier. Fig. 1 shows an overview of the trust triangle.
- Digital wallets are the digital equivalent of physical wallets, which can be held by advanced processing gadgets like laptops, smartphones, or tablets to keep verifiable credentials.
- Digital agents are the software modules that allow people to use their digital wallets. It can present and obtain verifiable credentials, control connections, safely communicate, and substitute verifiable credentials with other digital agents.
- Decentralized Identifiers (DIDs) are digital addresses powered by cryptography. It allows for the creating of novel, private, and secure shared associations between two parties. It does not require any centralized registration authority.
- Blockchain and verifiable data registries are shared databases that are safeguarded by cryptography and can act as a foundation of reliability for public keys and DIDs without being coordinated to attack or a single point of failure.
- Governance frameworks issued by governance authorities are the group of legal, technical, and business rules for utilizing SSI infrastructure, which will permit interactive digital trust environments of any volume.

3.3. Decentralized key management system (DKMS)

The decentralized key management system is a cryptographic technique without a centralized authority. This technique aids in guaranteeing that no third party can destroy the overall system's security and integrity. Participants can utilize this system to securely verify each other and endorse keys and certificates [48]. In Public Key Architecture (PKI), centralized or federated authorities issue public key certificates to consumers. On the other hand, with DKMS, the distributed ledger is the initial "root of trust" for all members that supports a new type of root identity called a decentralized identifier (DID). Again, DID is a uniform resource identifier (URI) or globally unique identifier for any participant (entity, individual, or thing), which is generated cryptographically [49]. As it is registered on a decentralized ledger, it does not need any centralized authority for its creation or registration. It allows for secure, unique, and private peer-to-peer connections between two parties. DIDs are associated with at least one pair of cryptographic keys: public and private keys. Since DIDs and their related public keys are stored in Blockchain, they consider secure and reliable dispersion of public keys, solving the middle-man attack. Overall, DKMS uses distributed ledgers' immutability, security,

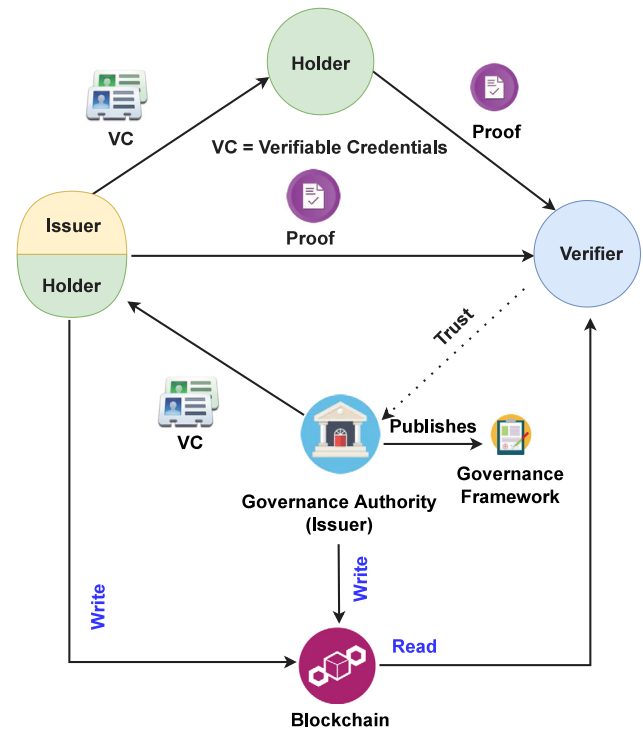


Fig. 1. Core roles in the trust triangle for verifiable credential.

availability, privacy, and resiliency properties to give highly scalable key dissemination, authentication, and recovery, finally making the PKI open to everybody.

3.4. Directed Acyclic Graph (DAG)

The Directed Acyclic Graph (DAG) is another type of distributed ledger technology [50]. A DAG is a network of separate transactions connected to numerous other transactions. DAG networks do not have transaction blocks. A DAG resembles a tree that grows from one transaction to another, to another, and so forth. Blockchain transaction histories are displayed as a "hashgraph" (a graph of Blockchain hashes) in DAG. Individual transactions offer validation for one another. Users of the network are both miners and validators; nevertheless, they cannot validate their own transactions. This often indicates that little to no costs are required in a DAG. Fantom Blockchain accomplishes its speed through DAG.

3.5. Lachesis consensus protocol

Lachesis is Fantom's asynchronous Byzantine fault-tolerant (aBFT) consensus algorithm, utilizing a DAG structure. Compared to Classical Byzantine Fault Tolerance (BFT), Practical Byzantine Fault Tolerance (pBFT), and Nakamoto consensus, Lachesis offers superior speed, security, and scalability. It permits participants to process commands without assuming specific roles, ensuring it functions smoothly even when up to a third of transaction data is fraudulent. Lachesis generates instantly usable output, confirming transactions within 1-2 s without the need for block confirmations. Consequently, Lachesis is renowned for its asynchronous, leaderless, and final attributes while maintaining Byzantine fault tolerance. This, in turn, empowers Fantom to provide rapid transaction speeds, cost-effective transactions, and deterministic finality [51].

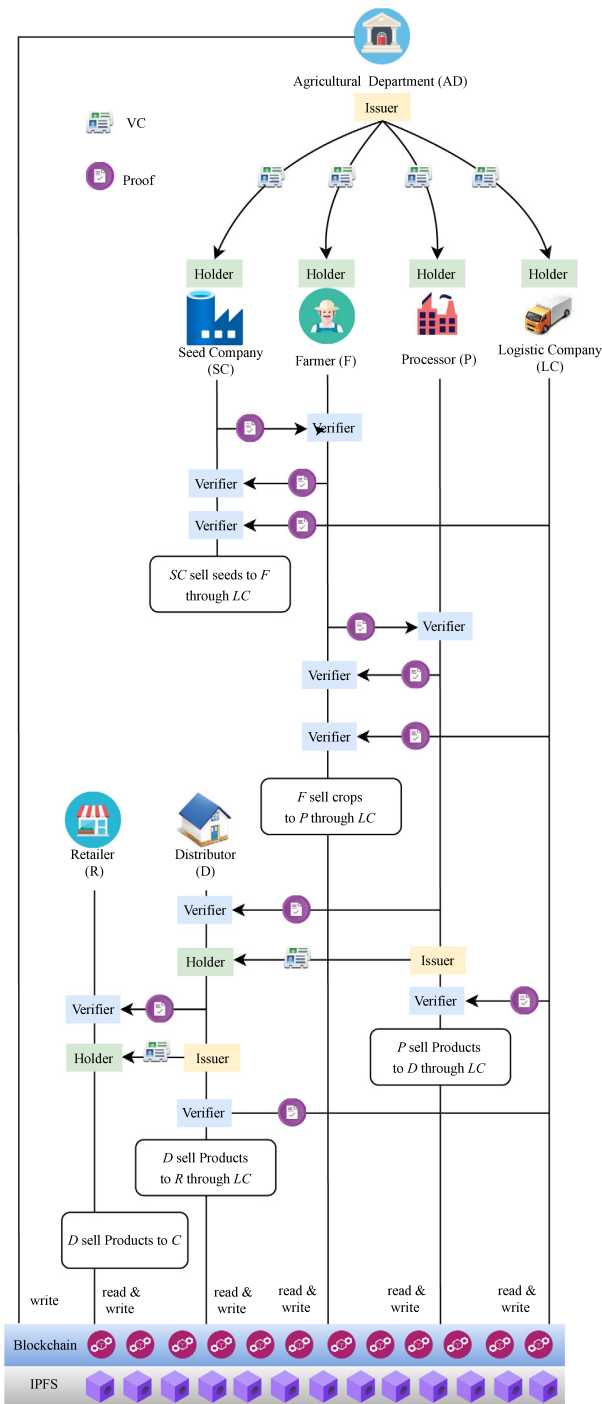


Fig. 2. Core roles in the system model and simple trust triangle for verifiable credential.

4. System overview

4.1. System model

This section briefly describes the Blockchain and SSI-based agricultural food and vegetable traceability system. Entities involved in this system are Seed Company (SC), Farmer (F), Processor (P), Distributor (D), Retailer (R), Consumer (C), Agricultural Department (AD), and Logistic Company (LC). The overview of the proposed system is shown in Fig. 2.

Their roles are described as follows: **Seed Company** SC produces a large number of seeds and sells them to the F. **Farmer** F purchases the seeds from SC and produces a huge number of crops and sells them to the P. **Processor** P buys the crops from F and refines and examines the crops, extracts the crops' supplementary materials, and converts the crops into finalized products. **Distributor** D purchases the final product from P and sells it to R. **Retailer** R buys the product, usually in batches from D and sells it to the C in a small quantity. **Customer** C are the end user who purchases the product from the R. **Agricultural Department** AD is a government authority that is responsible for issuing VC to SC, F, P, and LC. **Logistic Company** LC is responsible for delivering the product from seller to buyer. Generally, there are many SC, F, P, D, R, C, and LC but only one AD. The AD issues a VC to SC, F, P and LC. Besides this, P and D issue a VC to D and R respectively. According to this, SC, F, P, D, R, and LC work as a Holder H in the system. With reference to this, SC, F, P, D, R and C works as a verifier V of the VC's of the H's in the proposed system.

Participants AD, SC, F, P, D, R, C, and LC engage in mutual authentication through the execution of SSI-based entity authentication processes. In this SSI-based entity authentication procedure, I issues a VC to H, which is subsequently verified by V in accordance with H's consent. Each of I, H, and V follows the same protocol. Fig. 2 provides a comprehensive overview of the SSI-based entity authentication process. Following successful authentication, all entities participate in the traceability service, ensuring the traceability of agricultural products. The traceability data of each supply entity is directly recorded within the Blockchain using the current storage mode of the Blockchain traceability system. Consequently, the volume of transaction data grows in proportion to the number of entities, resulting in increased storage demands on the Blockchain and reduced query efficiency. To address these issues, we have implemented a dual storage system consisting of a decentralized file system, namely IPFS and the Blockchain. Further elaboration on the traceability service can be found in Section 5.

4.2. System setup

This section focuses on the system setup. Here, each entity of the supply chain has a decentralized identifier DID, associated with a pair of cryptographic keys: a public key PK and a private key SK. The public and private key pair is generated using Elliptic Curve Cryptography (ECC). The Elliptic Curve Digital Signature Algorithm (ECDSA) generates the digital signature. The system initially generates the DID, PK and SK for AD. DID_{AD} has only one associated DID document where DID_{AD} , PK_{AD} , and other necessary information is stored. The DID document of AD is encrypted and stored in IPFS. Following that, AD stores the hash of the encrypted DID document of AD into the Blockchain.

4.3. Issuance of VC

The VC issuance by the issuers is described in this section. All issuers follow the same protocol for issuing the VC. Here the VC of SC issuance by AD is explained as an example scenario. The Agricultural department AD creates DID_{SC} for the seed company SC. The DID_{SC} is VC of SC. Then, AD generates PK_{SC} , SK_{SC} against DID_{SC} . The DID_{SC} has an associated DID document where DID_{SC} , PK_{SC} , SK_{SC} and other necessary information will be stored. AD encrypts the DID document of SC and stores in IPFS. Following that, AD stores the hash of the encrypted DID document of SC into the Blockchain. AD uses SK_{AD} to digitally sign the VC, which it issues to SC. SC stores the VC in its digital wallet.

4.4. Verification of VC

The verification of VC is described in this section. Any verifier can verify any holder's VC upon the holder's approval. All verification process follows the same protocol. Here, the scenario of F wants to verify the VC_{SC} of SC is considered as an example. F requests for a digital proof of VC to the SC. If SC consents, SC's wallet generates proof and returns it to F. Since, the proof contains DID_{AD}, F uses it to read PK_{AD} and other cryptographic data from Blockchain. F uses DID_{AD} to verify that the proofs are valid and that the digital credential has not been tampered with.

4.5. Threat model and security goals

This study focuses on the traceability of agricultural products, but a few threats must be focused on. The two probable threats are: 1. Any entity may try to commit identity fraud to hide their crimes. 2. Any entity may try to infer the private data of other entities from IPFS. The security goals are: 1. No entity can commit identity fraud. 2. No entity can successfully infer the private data of another entity from IPFS.

5. Traceability service

The traceability service collects seeds from the SC and ends at the C's kitchen. The system includes six significant entities, i.e., seed company SC, farmer F, processor P, distributor D, retailer R, and consumer C. This section explains the working procedure of each element of the trusted traceability system.

5.1. Working method of SC

SC produces a huge number of seeds. These seeds will be identified by a unique QR code. SC packages the seeds according to the following four levels of packaging.

1. **Primary Packaging (Unit Level):** The product packaging level that touches the product itself, i.e., a crisp pack, a can, bags (e.g., paper, perforated polyethylene or polypropylene film, plastic or cotton nets).
2. **Secondary Packaging (Box Level):** The product packaging level that contains one or more primary packages, i.e., boxes or trays.
3. **Tertiary Packaging (Case):** The product packaging level that contains one or more secondary packages, i.e., case.
4. **Pallet :** Multiple tertiary packages are packed into a pallet for transport.

SC initially defines the number of units, boxes, cases, and pallets in which the seeds will be packaged. It also defines which unit will go to which box, which box to go to which case, and which case will go to which pallet. It will generate the unique QR code for each pallet, case, box, and unit, respectively. The unique QR Code generation process has four parts. They are: **Pallet:** $PA_{QRCode} = h(Comp_{Name}, PA_{SN})$, where, PA, h(), $Comp_{Name}$, and SN represent Pallet, hash function, company name, and serial number. Now, **Case:** $CA_{QRCode} = h(PA_{QRCode}, CA_{SN})$, where CA represent cases. Again, **Box:** $B_{QRCode} = h(CA_{QRCode}, B_{SN})$, where B represent Box. Further, **Unit product's:** $U_{QRCode} = h(B_{QRCode}, U_{SN})$, where, U represent Unit products.

1. **Pallet QR Code:** Each pallet has a unique QR code PA_{QRCode} and unique serial number PA_{SN} . The PA_{QRCode} is the hash of Company Name $Comp_{Name}$ and PA_{SN} . SC stores the raw data of seed such as country code, company type, company name, product type, batch no, and pallet package level in IPFS. Then, IPFS returns a transaction hash $IPFS_{PA}$. The SC stores PA_{QRCode} and $IPFS_{PA}$ stores in the QR Code Registry as a key-value pair.

2. **Case QR Code:** Each case has a unique QR code CA_{QRCode} and a unique serial number CA_{SN} . The QR code is the hash of PA_{QRCode} and case serial number CA_{SN} . SC stores the case package level as the raw data of seed in IPFS. Then, IPFS returns a transaction hash $IPFS_{CA}$. SC stores PA_{QRCode} and $IPFS_{CA}$ in the QR Code Registry as a key-value pair.
3. **Box QR Code:** Each box has a unique QR code B_{QRCode} and a unique serial number B_{SN} . The QR code is the hash of Case QR Code CA_{QRCode} and box serial number B_{SN} . SC stores the box package level as the raw data of seed in IPFS. Then, IPFS returns a transaction hash $IPFS_B$. SC stores B_{QRCode} and $IPFS_B$ in the QR Code Registry as a key-value pair.
4. **Unit QR Code:** Each unit has a unique QR code U_{QRCode} and a unique serial number U_{SN} . The QR code is the hash of Box QR Code (B_{QRCode}) and the unit serial number U_{SN} . SC stores the unit package level as the raw data of seed in IPFS. Then, IPFS returns a transaction hash $IPFS_U$. SC stores B_{QRCode} and $IPFS_U$ in the QR Code Registry as a key-value pair.

Then, SC packages the seeds from unit level to pallets and sells the seeds to F in pallets. At the time of selling, SC stores PA_{QRCode} and the selling information (i.e., the identity of the seller, selling time) as a key-value pair in the Seller Registry. The transaction data is stored in the Seller Registry.

5.2. Working process of F

F buys the seeds from the seed company. F usually buys the pallets from the SC. At the time of purchasing, F scans the unique QR code of the pallet using the verification application, verifies the authenticity of the seed, and traces the origin of the seeds. Algorithm 1 shows the seed verification process by F, where F scans the unique QR code of the pallet. The smart contract compares the scanned data with the keys stored in the QR Code registry. If it matches, then the verification application displays a message that the seeds are authentic. Otherwise, it displays a message that the seeds are fake. The system retrieves the value ($IPFS_{PA}$) of the key. The smart contract also compares the scanned data with the keys stored in the Seller registry. If it matches, then the verification application displays selling information (e.g., seller identity, selling time). The system compares $IPFS_{PA}$ with the transaction hashes returned by IPFS. If it matches, then the verification application displays the details of the seed. Thus, F can easily verify the authenticity of the seeds, trace the seeds, and easily know whether the seeds are already sold or not.

Algorithm 1: Verification of Seeds By Farmer.

```

1 F's Input: Scanned Data
2 System's Input:  $IPFS_{PA}$ 
3 Output: Message, Pallet Serial Seed's Details, Selling
    Information
4 if Scanned Data == Key in QR Code registry then
5   Returns Message = Authentic
6   Retrieves  $\{PA_{QRCode}, IPFS_{PA}\}$  from QR Code registry
7 end
8 else
9   Returns Message = Fake
10 end
11 if Scanned Data == Key in Seller registry then
12   Returns Selling Information = SellerID; Timestamp
13 end
14 if  $IPFS_{PA}$  == transaction hashes by IPFS then
15   Returns Seed's Details
16 end

```

After purchasing the seeds from SC, F cultivates the crops by using those seeds. Then, F stores the crops in the elevator after checking

relevant factors such as temperature, moisture, and time, which causes changes in grains during storage, as heat and humidity lead to quality deterioration, resulting in production loss. Following that, F sells the grains to P.

5.3. Working procedure of P

P purchases the grain from F. P refines the grains and analyzes the quality of grains. Afterward, P eliminates the moisture from the grains and makes the final product. Based on the order of D, P packages the product according to the following way. The product is packaged in a carton. A carton contains a product having a certain weight (e.g., a carton may contain 10 kg apples). After that, P generates the unique QR code for each carton. The unique QR code generation process is described below.

Each carton has a unique QR Code, P_{CART_QRCode} and a carton serial number P_{CART_SN} . The P_{CART_QRCode} is the hash of company name $Comp_{Name}$ and P_{CART_SN} . The unique QR Code generation process is $P_{CART_QRCode} = h(Comp_{Name}; P_{CART_SN})$. P stores the raw data of the product, such as country code, company type, company name, product name, product type, batch no., and package level of P in IPFS. IPFS returns a transaction hash $IPFS_P$. Then, P stores P_{CART_QRCode} and $IPFS_P$ in the QR Code Registry key-value pair. Besides this, P stores (P_{CART_QRCode}), selling information (i.e., the identity of the seller, selling time) as a key-value pair in the Seller Registry. Later, P packages the product in the carton and sends the product to D through the LC.

5.4. Working formula of D

D purchases the final product from P. D usually receives the product in a carton from P. At the time of purchasing or receiving, D can verify the authenticity of the product and trace the product by using the product verification application. The product verification process by D follows the same protocol as algorithm 1. D scans the unique QR code of the carton. The smart contract compares the scanned data with the keys stored in the QR Code registry. If it matches, then the verification application displays a message that the product is authentic. Otherwise, it displays a message that the product is fake. The system retrieves the value ($IPFS_P$) of the key. The smart contract also compares the scanned data with the keys stored in the Seller registry. If it matches, then the verification application displays selling information (e.g., seller identity, selling time). The system compares $IPFS_P$ with the transaction hashes stored in IPFS. If it matches, then the verification application displays the details of the product. Thus, D can easily verify the authenticity of the product, trace the product, and know whether a product is already sold. Thereafter, D sells the product to R. Based on the order of R, D packages the product according to the product packaging technique mentioned in Section 5.3. Then, D generates a unique QR code for each carton. The unique QR code generation process is described below.

Each carton of D has a unique QR Code D_{CART_QRCode} and a cartoon serial number D_{CART_SN} . The D_{CART_QRCode} is the hash of P_{CART_QRCode} and D_{CART_SN} . D stores the raw data of the product, such as country code, company type, company name, product name, product type, batch no., and package level of D in IPFS. IPFS returns a transaction hash $IPFS_D$. Then, D stores D_{CART_QRCode} and $IPFS_D$ in the QR Code Registry key-value pair. Besides this, D stores D_{CART_QRCode} , selling information (i.e., the identity of the seller, selling time, and transaction hash $IPFS_D$) as a key-value pair in the Seller Registry. Later, D packages the product in the carton and sends the product to R through the LC.

5.5. Working technique of R

R purchases the final product from D. At the time of purchasing or receiving, R verifies the authenticity of the product and traces the product by using the product verification application. The product verification process by R follows the same protocol as algorithm 1. R scans the unique QR code of the carton. The smart contract compares the scanned data with the keys stored in the QR Code registry. If it matches, then the verification application displays a message that the product is authentic. Otherwise, it displays a message that the product is fake. The smart contract retrieves the value ($IPFS_D$) of the key. The scanned data is basically D_{CART_QRCode} and as it is the hash of P_{CART_QRCode} and D_{CART_SN} , the smart contract retrieves the key-value pair $\{P_{CART_QRCode}; IPFS_P\}$ from QR Code registry. Besides this, the smart contract also compares the scanned data with the keys stored in the Seller registry. If it matches, then the verification application displays selling information (e.g., seller identity, selling time). The system separately compares $IPFS_D$ and $IPFS_P$ with the transaction hashes returned by IPFS. If it matches, then the verification application displays the details of the product by D and P, respectively. Thus, R can easily verify the authenticity of the product, trace the product, and know whether a product is already sold. Thereafter, R sells the product to C. Based on the order of C, R packages the product according to the product packaging technique mentioned in Section 5.3. Then, R generates a unique QR code for each carton. Then, R generates a unique QR code for each carton.

Each carton of R has a unique QR Code R_{CART_QRCode} and a cartoon serial number R_{CART_SN} . The R_{CART_QRCode} is the hash of D_{CART_QRCode} and R_{CART_SN} . R stores the raw data of the product, such as country code, company type, company name, product name, product type, batch no., and package level of R in IPFS. IPFS returns a transaction hash $IPFS_R$. Then, R stores R_{CART_QRCode} and $IPFS_R$ in the QR Code Registry as a key-value pair. Besides this, R stores D_{CART_QRCode} , selling information (i.e., the identity of the seller, selling time) as a key-value pair in the Seller Registry. Later, R sends the product to C through the LC.

5.6. Working mechanism of C

C purchases the final product from R. At the time of purchasing, C verifies the authenticity of the product and traces the product by using the product verification application. Algorithm 2 shows the product verification process by the retailer, where the retailer scans the unique QR code of the carton. The smart contract compares the scanned data with the keys stored in the QR Code registry. If it matches, then the verification application displays a message that the product is authentic. Otherwise, it displays a message that the product is fake. The scanned data is basically R_{CART_QRCode} and it is the hash of D_{CART_QRCode} and R_{CART_SN} , the smart contract retrieves the key-value pair $\{R_{CART_QRCode}; IPFS_R\}$ from QR Code registry. The smart contract also retrieves the key-value pair $\{D_{CART_QRCode}; IPFS_D\}$ from QR Code registry as R_{CART_QRCode} is the hash of D_{CART_QRCode} and R_{CART_SN} . The smart contract also retrieves the key-value pair $\{P_{CART_QRCode}; IPFS_P\}$ from QR Code registry as D_{CART_QRCode} is the hash of P_{CART_QRCode} and D_{CART_SN} .

Again, the smart contract also compares the scanned data with the keys stored in the Seller registry. If it matches, then the verification application displays selling information (e.g., seller identity and selling time). The system separately compares $IPFS_R$, $IPFS_D$ and $IPFS_P$ with transaction hashes returned by IPFS. If it matches, then the verification application displays the details of the product by R, D, and P. Thus, C can easily verify the authenticity of the product, trace the product, and know whether a product is already sold. Thus, in this way, from SC to C, each supply chain entity easily verifies, traces the agricultural product, and also knows whether the agricultural product is already sold or not in the supply chain, which ensures the traceability of agricultural products.

Algorithm 2: Verification By C.

```

1 C's Input: Scanned Data
2 System's Input: IPFSR, IPFSD, IPFSP
3 Output: Message, Product Details, Selling Information
4 if Scanned Data == key in QR Code registry then
5   Returns Message = Authentic
6   Retrieves {RCART_QRCode; IPFSR} from QR Code registry
7   Retrieves {DCART_QRCode; IPFSD} from QR Code registry
   As RCART_QRCode == h(DCART_QRCode, RCART_SN) Retrieves
   {PCART_QRCode; IPFSP} from QR Code registry As
   DCART_QRCode == h(PCART_QRCode, DCART_SN)
8 end
9 else
10  Returns Message = Fake
11 end
12 if Scanned Data == key in Seller registry then
13  Returns Selling Information = SellerID; Timestamp
14 end
15 if IPFSR == transaction hashes by IPFS then
16  Returns Product Details by R
17 end
18 if IPFSD == transaction hashes by IPFS then
19  Returns Product Details by D
20 end
21 if IPFSP == transaction hashes by IPFS then
22  Returns Product Details by P
23 end

```

5.7. Security analysis

Based on the threat model and security goals of Section 4.5, the security analysis has been done in this section. The proposed system consists of many entities that authenticate each other with SSI. The participants cannot be successful in doing identity fraud as SSI has the following properties.

- **User-Centric Control:** SSI places individuals at the center of their digital identity. It allows users to create, manage, and share their identity attributes through digital wallets. This means that individuals have the ultimate authority over their identity data, reducing the chances of third-party misuse or theft.
- **Decentralization:** SSI systems are typically built on decentralized, blockchain-based, or distributed ledger technologies. This decentralization means no central repository of identity data can be breached or hacked on a large scale. Instead, identity data is distributed and controlled by the individuals themselves.
- **Verifiable Claims:** SSI relies on the concept of verifiable claims, where third parties issue verifiable digital statements about individuals' attributes (e.g., age, qualifications, or employment history). These claims can be cryptographically signed and easily shared when needed. This reduces reliance on centrally stored databases and minimizes the exposure of personal information.
- **Selective Disclosure:** SSI allows individuals to share only the specific pieces of identity information required for a particular transaction or interaction. This principle of selective disclosure ensures that individuals do not need to reveal more information than necessary, reducing the risk of overexposing their identity.
- **Digital Signatures and Cryptography:** SSI employs strong cryptographic techniques, including digital signatures, to ensure the authenticity and integrity of identity claims. This makes it extremely difficult for malicious actors to forge or tamper with identity data.

- **Immutable Ledger:** The use of blockchain or distributed ledger technology ensures the immutability of identity records. Once a verifiable claim or identity attribute is recorded on the blockchain, altering or deleting it without the proper authorization becomes extremely difficult.
- **Revocation and Expiry:** SSI systems can support the revocation and expiry of verifiable claims. If a claim is no longer valid (e.g., a professional certification expires), it can be revoked, rendering it useless for fraudulent purposes.

While SSI offers significant advantages in reducing identity fraud risks, it is important to note that no system is entirely immune to all threats. SSI can greatly enhance security and user control over identity data, but its effectiveness also depends on proper implementation, user education, and ongoing security practices. Additionally, it does not protect against fraud that occurs outside the digital realm, such as physical document forgery.

On the other hand, any entity will also fail to breach the data privacy from IPFS as has the following features.

- **Content Addressing:** IPFS uses content addressing to locate and retrieve data. Instead of identifying files by their location or name, files are addressed by their content's cryptographic hash. This means that the same content will always have the same address (hash), regardless of where it is stored or who is accessing it. This feature helps protect data privacy by reducing the risk of data leakage or tracking through file names or URLs.
- **Decentralization:** IPFS operates on a decentralized network where files are distributed across multiple nodes. This decentralization ensures that there is no central authority or single point of control. Consequently, it becomes more challenging for third parties to monitor or control the distribution of data, enhancing user privacy.
- **Encryption:** While IPFS itself does not provide native encryption for data at rest, it allows users to encrypt their data before adding it to the IPFS network. By encrypting data before storage, users can ensure that only authorized parties with the decryption key can access and read the content.
- **Access Control:** IPFS allows users to implement access control mechanisms to restrict who can read or write data on the network. Users can define who can access their content and enforce these permissions when sharing data. This feature helps protect sensitive information from unauthorized access.
- **Private Networks:** Users can set up private IPFS networks, also known as "darknets", to share data within a closed group of trusted peers. These private networks are not accessible to the broader IPFS network, providing additional privacy and control.

In summary, IPFS provides several privacy-enhancing features through content addressing, decentralization, encryption, and access control, which can protect data privacy.

6. Experimental analysis

This section provides a description of the test equipment and examines the performance evaluation of the proposed model.

6.1. Testbed

Fantom Blockchain network tested machine configuration: Processor 2.3 GHz 8-core Intel Core i9, Turbo Boost up to 4.8G. Memory 16 GB of 2666 MHz DDR4. Graphics 2.6 GHz AMD Radeon Pro 5300M with 4 GB of GDDR6 memory

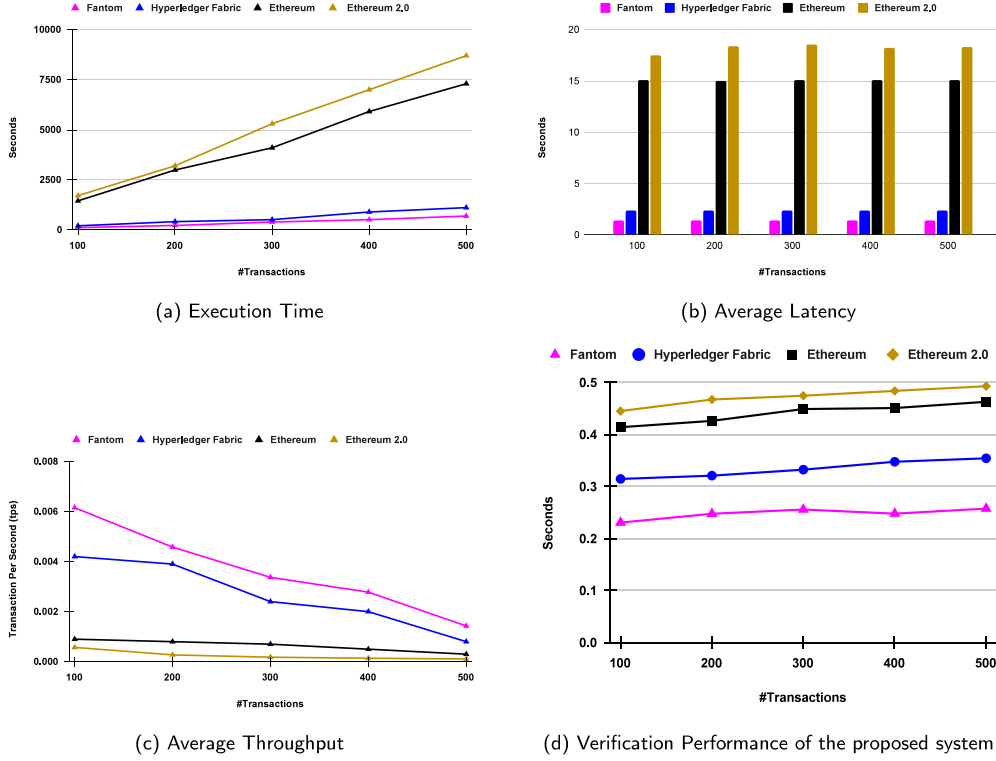


Fig. 3. Performance analysis of Fantom, Hyperledger Fabric, Ethereum and Ethereum 2.0.

6.2. Evaluation metrics

The results of the proposed framework are evaluated based on execution time (ET), average latency (AL), average throughput (AT), and average verification time (VT).

- ET: The total amount of time (in seconds) taken by a system to perform all the transactions, which is shown in Eq. (1) where N is the total number of transactions.

$$ET = \sum_{i=1}^N (T_2 - T_1) \quad (1)$$

T_1 and T_2 represent the transaction deployment time and transaction end time, respectively.

- AL: The average latency is the average of the difference between T_2 and T_1 , for a set of transactions, which is shown in Eq. (2).

$$AL = \frac{\sum_{i=1}^N (T_2 - T_1)}{N} \quad (2)$$

- AT: The average throughput is the average of successful transactions number per second over the execution time, which is shown in Eq. (3).

$$AT = \frac{\sum_{i=1}^N (T_2 - T_1)}{N} \quad (3)$$

6.3. Result evaluation

This section demonstrates the result analysis of the system. The proposed system is evaluated in execution time, average latency, and average throughput. Fig. 3 illustrates the performance analysis of Fantom, Hyperledger Fabric, Ethereum, and Ethereum 2.0.

This study investigates the variation in performance time consumption by changing the number of transactions in Fig. 3 while using four different Blockchain technologies: Fantom, Hyperledger Fabric,

Ethereum, and Ethereum 2.0. The x-axis displays the number of transactions ranging from 1 to 500, and the y-axis displays the total time consumption for various sets of transactions in seconds. The graph is represented on a linear scale. The execution time is proportional to the number of transactions. In the scenario, Ethereum hardly completes 980 transactions. Analysis shows that Fantom constantly consumes less time than Hyperledger Fabric, Ethereum, and Ethereum 2.0. The difference between Fantom, Hyperledger Fabric, Ethereum, and Ethereum 2.0 in execution time grows larger as the volume of transactions rises. In Fig. 3b and Fig. 3c, we evaluated the latency and throughput by varying the number of transactions with Fantom, Hyperledger Fabric, Ethereum, and Ethereum 2.0. The x-axis of both figures shows the transactions number, which varies from 1 to 500. The y-axis of Fig. 3b shows the average latency in seconds for every group of transactions. However, on that same axis, Fig. 3c shows the average throughput in transaction per second (tps) for individual transaction sets. Analysis of the performance reveals that the latency of Fantom is constantly lower, and throughput is constantly higher compared to Hyperledger Fabric, Ethereum, and Ethereum 2.0. Therefore, it proves that Fantom is faster than Hyperledger Fabric, Ethereum, and Ethereum 2.0. In summary, the proposed system provides more reliable performance in Fantom than Ethereum, Ethereum 2.0, and Hyperledger Fabric in terms of speed, cost, scalability, and security because of the DAG-based asynchronous byzantine fault tolerance consensus algorithm Lachesis.

In Fig. 3d, we evaluated the average verification performance of the proposed system by changing the number of transactions while using four different Blockchain technologies: Fantom, Hyperledger Fabric, Ethereum, and Ethereum 2.0. The x-axis shows the transaction number, which varies from 1 to 500. The y-axis displays the average verification time. Verification time is the time to verify all the credentials (i.e., VC of a supply chain entity, agricultural products) of the proposed system. The figure shows that the average verification time of Fantom is lower than Ethereum, Ethereum 2.0, and Hyperledger Fabric. On the other hand, the average verification performance of Ethereum and Ethereum 2.0 is the same. Therefore, it proves that Fantom is faster than Hyperledger Fabric, Ethereum, and Ethereum 2.0. In summary, the proposed

system gives a faster and more reliable performance in the Fantom Blockchain network than Ethereum, Ethereum 2.0, and Hyperledger Fabric in terms of average verification time.

7. Conclusion

We proposed a peer-to-peer supply chain network for trusted agricultural product traceability in this research. The study demonstrated that SSI, DKMS, and Fantom-based public Blockchain networks provide a faster, cheaper, and more reliable supply chain service. DKMS ensures a robust and trusted authentication service so that SSI becomes more reliable and distributed for each supply chain entity. Moreover, the proposed system is deployed on different public networks, i.e., Fantom, Ethereum, and Ethereum 2.0, and a private network, i.e., Hyperledger Fabric, and illustrates that the proposed Fantom-based Blockchain network outperforms in terms of transaction and verification timing. Fantom keeps the network faster regarding the number of transactions and verification compared with Ethereum and Hyperledger fabric, making the proposed traceability service more realistic for the agricultural supply chain integration.

In future research, we will integrate IoT devices at the edge with on-board machine learning and image processing mechanisms to automate the total agricultural supply chain system.

CRediT authorship contribution statement

A.S.M. Touhidul Hasan: Conceptualization, Methodology, Software. **Apubra Daria:** Data curation, Writing – original draft. **Rakib Ul Haque:** Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

- [1] F. R. O. for Asia & the Pacific, implementation of the global strategy in Bangladesh, 2022, URL <https://www.fao.org/asiapacific/perspectives/agricultural-statistics/global-strategy/results-in-the-region/bangladesh/en>. (Accessed 6 November 2022).
- [2] G. Francois, V. Fabrice, M. Didier, Traceability of fruits and vegetables, *Phytochemistry* 173 (2020) 112291.
- [3] S. Islam, L. Manning, J.M. Cullen, Systematic assessment of food traceability information loss: A case study of the Bangladesh export shrimp supply chain, *Food Control* 142 (2022) 109257.
- [4] A. Bushra, H.M. Zakir, S. Sharmin, Q.F. Quadir, M.H. Rashid, M.S. Rahman, S. Mallick, Human health implications of trace metal contamination in topsoils and brinjal fruits harvested from a famous brinjal-producing area in Bangladesh, *Sci. Rep.* 12 (1) (2022) 14278.
- [5] S.T. Tasnim, M.A. Islam, R.J. Taifa, S. Mahbub, M.R.A. Rashid, Agri-food traceability using blockchain technology to ensure value chain management and fair pricing in Bangladesh, in: 2022 IEEE 8th World Forum on Internet of Things, WF-IoT, IEEE, 2022, pp. 1–6.
- [6] M.A. Khan, M.E. Hossain, A. Shahaab, I. Khan, ShrimpChain: A blockchain-based transparent and traceable framework to enhance the export potentiality of Bangladesh shrimp, *Smart Agric. Technol.* 2 (2022) 100041.
- [7] K.Z. Hossain, J. Xue, M.G. Rabbany, Consumers' willingness to pay for GLOBALG, AP certified chicken: Empirical evidence from a consumer survey in Bangladesh, *Food Control* 130 (2021) 108397.
- [8] B. Zaabar, O. Cheikhrouhou, F. Jamil, M. Ammi, M. Abid, HealthBlock: A secure blockchain-based healthcare data management system, *Comput. Netw.* 200 (2021) 108500.
- [9] S. Akter, F. Reza, M. Ahmed, Convergence of blockchain, k-medoids and homomorphic encryption for privacy-preserving biomedical data classification, *Internet Things Cyber-Phys. Syst.* 2 (2022) 99–110.
- [10] M. Usmonov, Identification and authentication, 2021, Scienceweb academic papers collection.
- [11] J.P. Qian, X.T. Yang, X.M. Wu, L. Zhao, B.L. Fan, B. Xing, A traceability system incorporating 2D barcode and RFID technology for wheat flour mills, *Comput. Electron. Agric.* 89 (2012) 76–85.
- [12] G. Zhao, H. Yu, G. Wang, Y. Sui, L. Zhang, Applied research of IOT and RFID technology in agricultural product traceability system, in: *Computer and Computing Technologies in Agriculture VIII: 8th IFIP WG 5.14 International Conference, CCTA 2014, Beijing, China, September (2014) 16–19*, Springer International Publishing, 2015, pp. 506–514, Revised Selected Papers 8.
- [13] Y. Dong, B. Ding, G. Zhang, G. Jin, X. Zhao, Quality and safety traceability system based on agricultural product supply chain, *Trans. Chin. Soc. Agric. Eng.* 32 (1) (2016) 280–285.
- [14] T. Surasak, N. Wattanavichien, C. Preuksakarn, S.C. Huang, Thai agriculture products traceability system using Blockchain and internet of things, *System* 14 (15) (2019).
- [15] Y. Dong, X. Zhang, J. Xu, W. Xiaoyi, K. Jianlei, S. Pengcheng, Blockchain-based traceability model for grains and oils whole supply chain, *Food Sci.* 41 (9) (2020) 30–36.
- [16] K. Salah, N. Nizamuddin, R. Jayaraman, M. Omar, Blockchain-based soybean traceability in agricultural supply chain, *IEEE Access* 7 (2019) 73295–73305.
- [17] M. Conti, Food traceability in fruit and vegetables supply chain, in: 2020 IEEE International Symposium on Circuits and Systems, ISCAS, IEEE, 2020, pp. 1–5.
- [18] A. Shahid, A. Almogren, N. Javaid, F.A. Al-Zahrani, M. Zuair, M. Alam, Blockchain-based agri-food supply chain: A complete solution, *IEEE Access* 8 (2020) 69230–69243.
- [19] Y.U. Chuntang, Z.H.A.N. Yongzhao, L.I. Zhiyuan, Using blockchain and smart contract for traceability in agricultural products supply chain, in: 2020 International Conference on Internet of Things and Intelligent Applications, ITIA, IEEE, 2020, pp. 1–5.
- [20] D. Bumblauskas, A. Mann, B. Dugan, J. Rittmer, A blockchain use case in food distribution: Do you know where your food has been? *Int. J. Inf. Manage.* 52 (2020) 102008.
- [21] L. Cocco, K. Mannaro, Blockchain in agri-food traceability systems: A model proposal for a typical Italian food product, in: 2021 IEEE International Conference on Software Analysis, Evolution and Reengineering, SANER, IEEE, 2021, pp. 669–678.
- [22] X. Yang, M. Li, H. Yu, M. Wang, D. Xu, C. Sun, A trusted blockchain-based traceability system for fruit and vegetable agricultural products, *IEEE Access* 9 (2021) 36282–36293.
- [23] L. Wang, L. Xu, Z. Zheng, S. Liu, X. Li, L. Cao, et al., Smart contract-based agricultural food supply chain traceability, *IEEE Access* 9 (2021) 9296–9307.
- [24] L. Cocco, R. Tonelli, M. Marchesi, Blockchain and self sovereign identity to support quality in the food supply chain, *Future Internet* 13 (12) (2021) 301.
- [25] A.T. Hasan, S. Sabah, R.U. Haque, A. Daria, A. Rasool, Q. Jiang, Towards convergence of iot and blockchain for secure supply chain transaction, *Symmetry* 14 (1) (2022) 64.
- [26] L. Marchesi, K. Mannaro, M. Marchesi, R. Tonelli, Automatic generation of ethereum-based smart contracts for agri-food traceability system, *IEEE Access* 10 (2022) 50363–50383.
- [27] S. Yadav, T.M. Choi, S. Luthra, A. Kumar, D. Garg, Using Internet of Things (IoT) in agri-food supply chains: a research framework for social good with network clustering analysis, *IEEE Trans. Eng. Manage.* 70 (3) (2022) 1215–1224.
- [28] B. Subashini, D. Hemavathi, Detecting the traceability issues in supply chain industries using blockchain technology, in: 2022 International Conference on Advances in Computing, Communication and Applied Informatics, ACCAI, IEEE, 2022, pp. 1–8.
- [29] J. Katti, P. Inde, P. Jadhav, T. Ghule, D. Deshmukh, G. Dharmale, A block-chain based approach for food supply chain management, in: 2022 13th International Conference on Computing Communication and Networking Technologies, ICCCNT, IEEE, 2022, pp. 1–5.
- [30] B. Subashini, D. Hemavathi, Blockchain and IoT's role in agri food supply chain traceability, in: 2022 IEEE 9th Uttar Pradesh Section International Conference on Electrical, Electronics and Computer Engineering, UPCON, IEEE, 2022, pp. 1–6.
- [31] Y. Zhang, G. Zhang, B. Feng, X. Chen, B. Wang, Q. Li, Flexible, safe and trusted agricultural product blockchain traceability system, in: 2022 International Conference on High Performance Big Data and Intelligent Systems, HDIS, IEEE, 2022, pp. 304–308.
- [32] G.S.R. Krishna, P. Rekha, Food supply chain traceability system using blockchain technology, in: 2022 8th International Conference on Signal Processing and Communication, ICSC, IEEE, 2022, pp. 370–375.
- [33] Y. Wang, X. Zhang, J. Leng, S. Cheng, D. Wang, Design of fruit and vegetable produce traceability system based on dual chain and dual storage blockchain, in: 2022 International Conference on Cloud Computing, Big Data and Internet of Things, 3CBIT, IEEE, 2022, pp. 130–134.
- [34] P. Saranya, R. Maheswari, Proof of transaction (PoTx) based traceability system for an agriculture supply chain, *IEEE Access* 11 (2023) 10623–10638.

